AKHIL YEDUVAKA

Academy of Computer Science • Morris County School of Technology • Denville, NJ
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GPA 4.0 / 4.3 • SAT 1370 • PSAT 1450 • National Honor Society • Expected Graduation: June 2027

PROFESSIONAL SUMMARY

Motivated Computer Science student with hands-on experience in full-stack and mobile development, Python and Java application development, hardware-software integration, robotics engineering, and cybersecurity. Proven ability to lead technical teams, ship functional software across web, mobile, and embedded platforms, and rapidly learn new technologies. Notable work includes a NASA App Development Challenge submission and a bionic glove therapeutic system. Seeking an internship in the summer or fall to apply to and expand skills in software development, data analytics, or engineering.

TECHNICAL SKILLS

Languages: Python • Java • C++ • JavaScript • HTML / CSS • R

Frameworks & Tools: Expo (React Native) • Arduino • Tkinter • Git / GitHub • VS Code • Godot • Replit • Onshape (CAD) • Visual Studio

Skills: Full-Stack Web Development • Mobile Development • Hardware-Software Integration • OOP • CTF / Cybersecurity • Robotics (FRC) • Data Analysis

Cybersecurity Tools: Wireshark • Steghide • CyberChef • Linux CLI • Metadata Analysis

PROJECTS

NASA App Development Challenge | *NASA / Multi-Platform System* 2025

- Engineered a multi-platform system to simulate the Apollo 11 flight plan, meeting NASA's specific technical visualization and data requirements.
- Built a mobile interface using Expo (React Native) to serve as the primary bridge between a Java-based flight simulation engine and a web-based information portal.
- Aligned a multi-platform architecture to connect the public-facing website, mobile interface, and Java flight simulation engine.
- Tools: Expo (React Native), Java, HTML / CSS, JavaScript.

NeuroMaker — Bionic Glove Therapy System | *Hardware-Software Integration Project* 2025

- Engineered a diagnostic and therapeutic solution using a bionic glove to help clinicians track hand mobility and optimize physical therapy for patients with motor impairments.
- Developed core control logic for the NeuroMaker Hand 2.0 in C++ (Arduino), mapping sensor data from the glove to measurable therapeutic information.
- Documented technical findings and sensor-to-result mappings to support clinical decision-making and iterative therapy adjustments.
- Tools: Arduino (C++), NeuroMaker Bionic Kit, Hardware-Software Integration.

Digi-Cure Healthcare App | *Congressional App Challenge* 2024 – 2025

- Designed and co-developed a full-stack healthcare desktop application in Python (Tkinter) that enables users to safely navigate over-the-counter medications.
- Engineered a searchable medication database filtering hundreds of drugs by symptom and brand, with integrated safety warnings and ingredient lists.
- Built an AI-powered chatbot module to answer user questions about medications, extending core app functionality.
- Managed project architecture, UI design, and end-to-end testing with a development partner.

Web Development Portfolio | *Personal Projects* 2023 – Present

- Developed multiple responsive, interactive websites using HTML, CSS, and JavaScript deployed via Vercel, CodeHS, and Replit.
- Projects include multi-page informational and interactive websites — demonstrating proficiency in layout, interactivity, and deployment workflows.

Console Applications | *Personal Projects* 2023 – Present

- Authored numerous standalone programs in Java, Python, and C++ covering diverse domains including: a classroom assignment tracker, Tic-Tac-Toe engine, coupon redemption tool, zoo management system, and playlist simulator.
- Applied object-oriented principles, data structures, and user-facing I/O design across all projects.

Game Development | *Personal Projects* 2024 – Present

- Built a 3D horror game and a 2D platformer using Godot Engine; created original assets with Blender (3D modeling) and custom pixel art.

LEADERSHIP & EXPERIENCE

Mechanical Subteam Leader | *MCST Robotics Club (FRC)* 2024 – Present

- Led a subteam in the full design-build-test cycle of a competition-ready FRC robot, ensuring all mechanical systems met strict technical specifications.
- Designed robot subsystems — including tank/swerve chassis, power transmissions, and a basic shooter — using Onshape CAD software.
- Collaborated with the programming subteam to integrate software and electronic hardware; diagnosed and resolved technical failures through iterative testing.
- Mentored junior members on safe shop tool usage and CAD workflows, improving team production speed and reducing assembly errors.

CTF Competitor | *GardenState CTF • PicoCTF* 2024 – 2025

- Competed in two Capture the Flag cybersecurity competitions, solving challenges in steganography, network analysis, and cryptography.
- Applied tools including Wireshark, Steghide, CyberChef, and Linux command-line utilities to decrypt ciphers and manipulate image/metadata files.

LeetCode Practitioner | *Self-Directed* 2024 – Present

- Regularly solves algorithmic and data-structure problems on LeetCode to strengthen problem-solving skills.

Project Lead — Cancer Walkathon | *Morris County Community Event* 2024

- Directed a team of peers using Agile-style task delegation to produce a high-impact promotional display, completing the project ahead of schedule.
- Led the team through feedback and revision sessions to produce a high-quality, illustrated trifold board used for community outreach and presentations.

EDUCATION

Academy of Computer Science | *Morris County School of Technology* Expected June 2027

- GPA: 4.0 / 4.3 | SAT: 1370 | PSAT: 1450 | National Honor Society
- Relevant Coursework: Introduction to Data Science • Software Design • Emerging Technologies • Introduction to Java Programming • Computer Science Programming I & II Honors